

JASPREET KAUR

Data Science & AI Engineer



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PROFESSIONAL SUMMARY

Recent B.Sc. Computer Science graduate from the University of Delhi and current M.Sc. Data Science student at Vellore Institute of Technology. 2.5 years of experience as a Web Development & Python Mentor at an EdTech startup. Build AI-driven and full-stack projects using Python, SQL, FastAPI, React, LangChain, LangGraph, and CrewAI, including RAG-based chatbots and agentic workflows. Interested in AI research, retrieval systems, and systems engineering.

EDUCATION

M.Sc. Data Science, Vellore Institute of Technology, India — 2025-Ongoing

Relevant coursework: Machine Learning, Deep Learning, Statistical Modeling, Data Engineering, NLP

B.Sc. Computer Science, University of Delhi, India — 2019-2020

Core topics: Algorithms, Data Structures, Databases, Operating Systems, Python Programming

EXPERIENCE

Web Development & Python Mentor, CodeYoung (Remote, India) — June 2023 – Present

- Designed and delivered Python, web development, and data curricula for school students with a project-based approach.
- Mentored 50+ individuals on building small web applications using React, Node.js, and FastAPI.
- Collaborated with curriculum and product teams to design mini-projects that reflect real engineering workflows.

PROJECTS

RAG-Powered Question Answering Chatbot — Python, FastAPI, React, LangChain, FAISS/Chroma

- Build an end-to-end RAG system that ingests documents, creates embeddings, and retrieves relevant content for question answering.
- Improved the backend in FastAPI and connected it to a React frontend.
- Tested chunking strategies, embedding models, and prompting approaches to improve response quality.
- Added logging to analyse query patterns and retrieval behavior.

GenAI and Agentic Workflows Exploration — Python, LangChain, LangGraph, and CrewAI

- Designed multi-step workflows for planning, retrieval, and tool use in AI agents.
- Build simple agent-based assistants for question answering and task planning.
- Gained familiarity with tools, state, memory, routing, and multi-agent orchestration.

MSc Data Science Mini-Project — [Project Name]

- Applied (regression/classification) for [dataset] using Python and scikit-learn, tuning hyperparameters and evaluating performance with standard metrics.

SKILLS

Programming & Web

- Python
- SQL
- JavaScript
- React
- Node.js
- FastAPI

ML & Data Science

- Supervised ML
- Data Preprocessing
- Feature Engineering
- Evaluation Metrics
- Basic NLP

AI & GenAI

- LLMs
- RAG
- LangChain
- LangGraph
- CrewAI
- Agentic Workflows

Tools

- Git
- Jupyter
- VS Code
- REST APIs
- JSON

ADDITIONAL

Ongoing focus on LLM research topics such as alignment, retrieval-augmented reasoning, and evaluation for research-engineer roles.